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On the distribution of trace element concentrations in multiple bone elements in 10 Danish medieval and post-medieval individuals

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Abstract

Background: The differences in trace element concentrations among 19 different bone elements procured from 10 archaeologically derived human skeletons have been investigated. The 10 individuals are dated archaeologically and some by radiocarbon dating to the medieval and post-medieval period, an interval from ca. AD 1150 to ca. AD 1810. This study is relevant for two reasons. First, most archaeometric studies analyze only one bone sample from each individual; so to what degree are the bones in the human body equal in trace element chemistry? Second, differences in turnover time of the bone elements makes the cortical tissues record the trace element concentrations in equilibrium with the blood stream over a longer time earlier in life than the trabecular. Therefore, any differences in trace element concentrations between the bone elements can yield what can be termed a *chemical life history* of the individual, revealing changes in diet, provenance, or medication throughout life.

Methods: Thorough decontamination and strict exclusion of non-viable data has secured a dataset of high quality. The measurements were carried out using Inductively Coupled Plasma Mass Spectrometry (for Fe, Mn, Al, Ca, Mg, Na, Ba, Sr, Zn, Pb and As) and Cold Vapor Atomic Absorption Spectroscopy (for Hg) on ca. 20 mg samples.

Results: Twelve major and trace elements have been measured on 19 bone elements from 10 different individuals interred at five cemeteries widely distributed in medieval and renaissance Denmark. The ranges of the concentrations of elements were: Na (2240–5660 $\mu\text{g g}^{-1}$), Mg (440–2490 $\mu\text{g g}^{-1}$), Al (9–2030 $\mu\text{g g}^{-1}$), Ca (22–36 wt. %), Mn (5–11450 $\mu\text{g g}^{-1}$), Fe (32–41850 $\mu\text{g g}^{-1}$), Zn (69–2610 $\mu\text{g g}^{-1}$), As (0.4–120 $\mu\text{g g}^{-1}$), Sr (101–815 $\mu\text{g g}^{-1}$), Ba (8–880 $\mu\text{g g}^{-1}$), Hg (7–78730 ng g^{-1}), and Pb (0.8–426 $\mu\text{g g}^{-1}$).

Conclusions: It is found that excess As is mainly of diagenetic origin. The results support that Ba and Sr concentrations are effective provenance or dietary indicators. Migrating behavior or changes in diet have been observed in four individuals; non-migratory or non-changing diet in six out of the 10 individuals studied. From the two most mobile (most changing diet) individuals in the study, it is deduced that the fastest turnover is seen in the trabecular tissues of the long bones and the hands and the feet, and that these bone elements have higher turnover rates than centrally placed trabecular bone tissue, such as from the ilium or the spine. Comparing Sr and published bone turnover times, it is concluded that the differences seen in Sr concentrations are not caused by diagenesis, but by changes of diet or provenance. Finally, it is concluded that there can be two viable inter-

pretations of the Pb concentrations, which can either be seen as an indicator for social class or a temporal development of increased Pb exposure over the centuries.

KEYWORDS

barium, Denmark, lead, multiple bone samples, social indicator, strontium, trace element analyses

1 | INTRODUCTION

Human bones procured from archeological excavations are often analyzed for trace elements or isotopes to reveal information pertinent to the interpretation of the life and death of single individuals and to the community they came from. However, in most studies, the analyses are limited to one sample from each individual, and only in relatively few cases have more samples from the same individual been analyzed (e.g., Carvalho, Marques, Lima, & Reus, 2004; Martinez-Garcia et al., 2005; Rasmussen et al., 2013; Rasmussen, Skytte, Jensen, & Boldsen, 2015; Rebocho, Carvalho, Marques, Ferreira, & Chettle, 2006; Skytte & Rasmussen, 2013). There can be many reasons for selecting only one sample from each individual, even though such thoughts are rarely made explicit: (1) the assumption that the bones are chemically alike; (2) one particular bone element or tooth is advantageous over others for the study in question; (3) curators arguments for the conservation of the material in general, and in connection with possible future exhibitions in particular; (4) to limit the number of analyses in the laboratory and/or maximize the number of individuals analyzed.

Which elements to analyze for, can also be selected for a variety of reasons. In archaeologically derived human bones, the elements can be selected according to at least three different criteria: (1) the element is controlled by diet, medication or environmental exposure; (2) the element is controlled by a metabolic mechanism; (3) diagenetic processes have been adding or removing the element from the bone tissue after death.

For the assessment of single adult individuals, it could be advantageous to select a bone sample that reflects the years of life shortly before death—where the individual played its most mature and significant role in society. This criterion is generally met by the trabecular tissues, which have fast turnover rates, and which reflects the chemistry in the body late in life. The cortical tissues have a slower turnover rate, and therefore reflect the chemistry at an earlier time in life. However, the cortical tissues have other distinct advantages. They are less prone to diagenesis and much easier to decontaminate by mechanically removing the surface (Rasmussen et al., 2013, 2015). Consequently, the cortical tissues are more useful for revealing minor differences in trace element concentrations, which are particularly important for the study of processes on the population level.

The goal of the present work is to elucidate similarities and differences in trace element concentrations within and between individuals. Furthermore, we try to ascertain from the trace element data the causes of the variations in trace element concentrations. In the present work, trace element concentration data from 19 different bone elements in 10 medieval and post-medieval individuals excavated in five

different cemeteries located at four different locations in Denmark are presented. This makes it possible to characterize the five cemeteries studied. Complex relations between the trace element concentrations of the different bone elements of single individuals are for the first time analyzed and reported.

2 | MATERIALS

Ten different skeletons, four females and six males aged from the late teens to around 60 years and suffering from different diseases at the time of death have been selected for this study. To cover widely different geographical areas, two skeletons were selected from each of five different cemeteries (see Table 1 for a summary). The five cemeteries are:

- 1 The Franciscan Friary of Svendborg was excavated by Svendborg Museum in 1975–1977 (see Kristensen, 1994 and references therein). Svendborg was a well-developed medieval town with an industrious harbor. The two selected individuals were buried in the laymen cemetery in the second layer of excavated graves (the grave locations can be seen in Kristensen, 1994, p. 87). The collagen fraction of AG18 has been radiocarbon dated to 515 ± 40 BP (KIA-50617, KLR-6574) with a $\delta^{13}\text{C}$ -value of -18.2 per mille VPDB, which gives rise to a marine food reservoir effect of 74 years (Rasmussen, 2008). This gives a calibrated date interval at ± 2 standard deviations of AD 1410–1520 and 1595–1620. This is in accordance with a dendrochronological date of the felling year of the wood in the coffin of AG18 of 1473–1474 (Kristensen, 1994). The cemetery reflects a highly mobile urban population.
- 2 The cemetery surrounding Sct. Nicolai Church is also situated in Svendborg. It was excavated in 1995 by Svendborg Museum (Thomsen, 2008). The individual A1 is from the post-medieval, and radiocarbon dated to 201 ± 21 (KIA-20907, KLR-5447), which based on the $\delta^{13}\text{C}$ -value of -18.6 per mille VPDB must be corrected for 54 years for marine reservoir effect. This yields a calibrated date interval at ± 2 standard deviations of AD 1670–1953 (Rasmussen, 2008). However, the cemetery went out of use in the first decade of the 19th century, which dates the skeleton to AD 1670–1810. Skeleton A11 is less well documented, but based on the archaeological evidence it can be referred to the medieval or early post medieval period. The cemetery reflected a highly mobile and urban population.
- 3 Tirup was a small rural village situated 1.3 km outside of the medieval town of Horsens. The usage of the cemetery has been dated to the period AD 1150–1350 based on archaeological and

TABLE 1 The skeletons analyzed. Information about the skeletal material included in the present investigation. Sex and age at death are based on experience based methods (Milner & Boldsen, 2012)

Skeleton	Site	Dating ^a	Sex	Age	Pathology	Environment	Burial type
AG18	Svendborg Franciscan	MP	M	50–55	Prostate cancer	Urban	Coffin
AG134	Svendborg Franciscan	MP/PM	F	38–48	None	Urban	No coffin
A1	Svendborg St. Nicolai	MP	F	18–22	Syphilis	Urban	Coffin
A11	Svendborg St. Nicolai	MP/MP	F	42–52	None	Urban	No record
G296	Tirup	MP	M	16–18	Bone infection	Rural	No coffin
G307	Tirup	MP	M	28–32	Possibly TB	Rural	No coffin
B	Ærø	MP	M	25–30	TB and wounds from sharp instruments	Rural	In church
W	Ærø	MP	M	55–65	TB	Rural	No coffin
K1029	Ribe	PM	F	37–47	Syphilis	Urban	Coffin
K1256	Ribe	MP/PM	M	45–55	None	Urban	No coffin

^aMP = Medieval Period, PM = Post Medieval Period.

documentary sources (Boldsen, Kieffer-Olsen, & Pentz, 1989; Kieffer-Olsen, Boldsen, & Pentz, 1986). Both skeletons analyzed were buried in arm position “A,” this and the location and stratigraphy of the burials on the cemetery indicates that they died in the 13th century (Kieffer-Olsen, 1993). The cemetery reflects a rural population with clearly lower mobility than seen in urban communities (Boldsen, 1989, 1985; Rasmussen et al., 2015).

- 4 The chapel of Sct. Albert, Sct. Alberts Kapel, is situated on the southern coastline of the island Ærø. The building complex included a church, a large tower, and the cemetery. The cemetery was excavated in 1995–1996 (Skårup, 1997). The cemetery is dated archaeologically to between AD 1250 and the reformation in AD 1536. A surprisingly large number of the men (5 out of 22) had unhealed lesions from sharp weapons. It is hard to characterize the population which is on one side situated on a fairly isolated island, but where many of the men saw an unusual amount of violent action (Rasmussen et al., 2015). The individual in grave W has been radiocarbon dated to 630 ± 20 (KIA-50600, KLR-10222), which based on the $\delta^{13}\text{C}$ -value of -19.2 per mille VPDB needed no correction for the marine reservoir effect. This yields a calibrated date interval at ± 2 standard deviations of AD 1298–1395.
- 5 Ribe is probably the oldest town in present day Denmark (Feveile, 2006; Feveile & Søvsø, 2010; Nielsen, 1985; Nyborg & Mulvad, 1988), holding the earliest Christian communities in Denmark. The investigated graves are from the cemetery *Lindegaarden* adjacent to the cathedral. The cemetery was excavated in 2008 and 2009 by Sydvestjyske Museer (excavation code ASR13). Skeleton K1029 was buried in the second youngest layer of the procession walk with a stratigraphic dating of ca. AD 1650–1700. The other individual, K1256, was from the courtyard of the procession walk and has a stratigraphy which dates it to the 1500s.

The geographical locations of the cemeteries are shown in Figure

1. All the skeletons are stored at the ADBOU (Unit of Anthropology, Department of Forensic Medicine, University of Southern Denmark).

A maximum of 19 bone samples have been collected from each individual, six samples of cortical bone and 13 samples of trabecular tissues. Cortical tissue has been sampled from right and left femur, left tibia, right and left humerus, and left ulna. Trabecular tissue has been sampled from a vertebra, sacrum, ilium, two ribs, right and left femur, left tibia, a tarsal bone, right and left humerus, and a carpal bone. The full set of samples is marked in the drawing of Figure 2. The reason samples have been selected from both the left and the right femora and humeri are to reveal if the chemistry of the skeleton is really

**FIGURE 1** Location map showing the positions of the four locations in the present study in Southern Denmark. Svendborg is represented by two different cemeteries, the laymen cemetery at the Franciscan Friary and the cemetery surrounding the parish church Sct. Nicolai

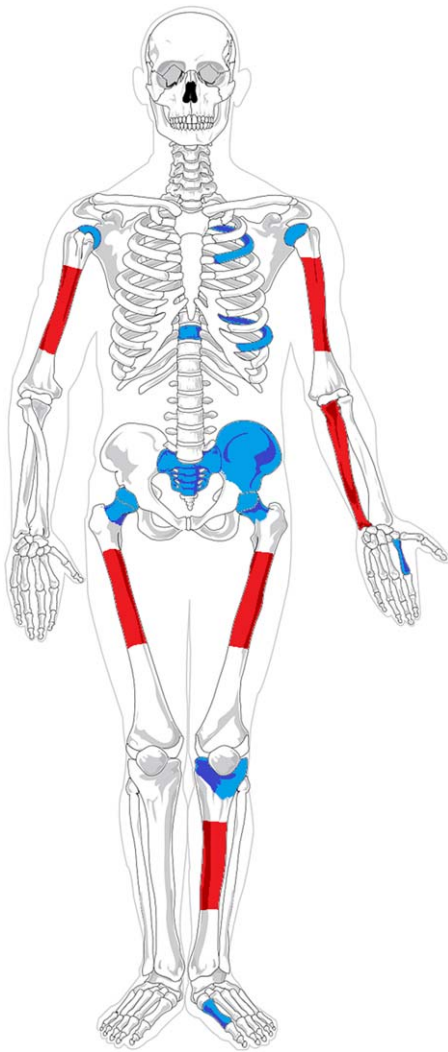


FIGURE 2 Bone elements sampled from the investigated individuals. Red areas represent places where cortical tissue has been sampled, and blue areas indicate where trabecular samples have been taken. Positions of ribs, vertebra, ilium, ulna, carpal, tibia, and tarsal may differ depending on availability and preservation

symmetric. It is possible that there could be systematic differences between left and right, for instance caused by side-specific prolonged work such as operating a potter's wheel. The analyzed individuals do not all include a full set of samples, as some bone elements may have been missing or excluded; either due to poor preservation or because the Ca-concentration turned out to be below 22.1 wt. %. The selected and the analyzed samples are listed in Table 2.

One limitation of the present project is the spatial resolution of the sampling. A total of about 40 mg of bone tissue was used for each analysis, but often 80–100 mg was sequestered while drilling the sample out. This represents the volume of about 100 mm³. Other techniques such as LA-ICP-MS have a much better spatial resolution (ca., 10 µm), and synchrotron radiation micro-XRF an even better resolution (ca., 1 µm). However, it is a lot harder to obtain an absolute calibration

in µg g⁻¹ for both these techniques than it is for solution-ICP-MS, rendering the analytical uncertainty significantly better for the solution-ICP-MS.

3 | METHODS

All the skeletons analyzed are part of the permanent collection at ADBOU. As such they have all been washed before sampling. The samples were cut out with a Dremel Multipro drill mounted with a stainless steel drill of 2 mm in diameter. The drill was washed in deionized water and heated to dryness in an ethanol flame between each drilling operation. The surface of the cortical bone material was removed by a pre-drilling, and only the material subsequently drilled out was used for analyses. This was done to avoid contamination from soil and from the post-excavation handling of the bone. For the trabecular tissue samples this cleaning procedure was not possible, as the trabecular bone material is made up of very small and thin lamellae. Most of the trabecular tissue has been protected from contamination introduced by handling, as it is situated inside a shell of cortical bone, but it was only rarely possible to obtain samples that have not been in contact with penetrating soil. However, an attempt to remove contaminants in the form of adhering particles of soil or decayed soft tissue have been carried out by means of mechanical cleaning, where the sample was placed under a binocular microscope and stainless steel tweezers were used to sort the material. All utensils used were pre-cleaned prior to each operation.

The samples were analyzed by Inductively Coupled Plasma Mass Spectrometry (ICP-MS) and on a dedicated Cold Vapor Atomic Absorption Spectrometer (CV-AAS) for Hg. (Skytte & Rasmussen, 2013; Rasmussen et al., 2015). From each sample about 40 mg was dissolved in a mixture of 4 mL of HNO₃, 2 mL of H₂O₂, and 0.5 mL of HCl, all of which were ICP-MS grade. The dissolution took place in sealed polystyrene containers, which were left on a shaking table for 24 hours at room temperature (ca., 20°C). The samples were then diluted to a volume of 10 mL with Milli-Q water and filtered through a 0.45 µm disposable filter. After filtering, half of the solution was used for ICP-MS, the other half for CV-AAS for Hg. The solutions were stored at +4°C until the analyses were performed. The analyses were carried out with a Bruker ICP-MS 820 spectrometer equipped with a frequency-matching RF generator and a skimmer gas Reaction System (CRI) operating with He or H₂ to minimize interferences by polyatomic species. The CRI Reaction System was activated for Cr-52, Ni-60, and As-75 using He as a skimmer gas, and for Fe-56 using H₂. In all analyses, Sc-45, Y-89, and Tb-159 were used as internal standards. The rest of the elements were analyzed without skimmer gas. Interfering masses have been corrected for in all relevant cases with an algorithm in the machine software. Multi-element calibration standards were prepared in 1% HNO₃ at six different concentrations (0, 1, 10, 20, 100, and 200 µg L⁻¹). For each element, only three of these standards were selected to fit the appropriate concentration range. The methodology used here is as described in Skytte and Rasmussen (2013) and Rasmussen et al. (2015).

TABLE 2 Actual samples taken: X: Sample present and analyzed; C: Sample analyzed but data excluded due to low Ca-value; blank: Sampling was not possible due to lack of bone element or poor preservation

Sample	Individual									
	Svendborg Friary		Svendborg Nicolai		Tirup		St. Albert		Ribe	
	AG18	AG134	A1	A11	G296	G307	B	W	K1029	K1256
Fem. comp. (R)	X	X	X	X	X	X	X	X	X	X
Fem. comp. (L)	X	X	X	X	X	X	X	X	X	X
Tibia comp.	X	X	X	X	X	X	X	X	X	
Hum. comp. (R)	X	X	X	X	X	X	X	X	X	X
Hum. comp. (L)	X	X	X	X	X	X	X	X	X	X
Ulna comp.	X	X	X	X	X	X	X	X	X	X
Vertebrae	X	C	C	C	X	X	X	X	X	X
Sacrum	X	X	X	X	X	X	X	C	X	X
Ilium	C	X	X	X	X	X	X	X	X	X
Rib - high	X	X	X	X	X	X	X	X	X	X
Rib - low	X	X	X	X	X	X	X	X	X	X
Fem. spong. (R)	X	X	X	C	X	X	X	C	X	X
Fem. spong. (L)	X	X	C	X	X	X	X	X	X	X
Tibia spong.	X	X	C	C	X	X	X	X	X	
Tarsal spong.	X	X	X	C	X	X	X	X		
Hum. spong. (R)	X	X	X		X	X	X	X		X
Hum. spong. (L)	X	X	X		X	X	X	X	X	X
Carpal spong.	X	X	X	X	X	X	X	X		

The concentrations of Fe, Mn, Al, Ca, Mg, Na, Ba, Sr, Zn, Pb, As, and Hg were measured quantitatively in each sample. A sample of the NIST standard reference material SRM-1486 "Bone Meal" and an in-house bone standard made of homogenized medieval cortical bone tissue were measured daily. Samples displaying a Ca-concentration of less than 22.1 wt. % were assessed to be significantly contaminated by soil or soft tissue and not reflecting the correct bone value and the data for such samples were discarded (see Table 2 and Rasmussen et al., 2015 for details on this criterion).

4 | RESULTS

The following elements were analyzed for and reported here: Fe, Mn, Al, Ca, Mg, Na, Ba, Sr, Zn, Pb, As, and Hg. The results of the analyses are given in Table 3 (Supporting Information) as concentrations $\pm$ one standard deviation. The limit of quantization (LOQ) was calculated for all elements based on ca. 6 months of analyses. If a concentration is found to be lower than the LOQ, the value is quoted as "< LOQ." The entire data set is depicted in Figure 3, where the red color signifies cortical tissue samples and the blue color trabecular. All the reported elements were largely above LOQ and their distributions can be understood one way or another as described below.

The ranges and medians of the concentrations of elements were: Na (2240–5660; 3620 $\mu\text{g g}^{-1}$), Mg (440–2490; 955 $\mu\text{g g}^{-1}$), Al (9–2030; 247 $\mu\text{g g}^{-1}$), Ca (22–36; 27 wt. %), Mn (5–11450; 476 $\mu\text{g g}^{-1}$), Fe (32–41850; 1495 $\mu\text{g g}^{-1}$), Zn (69–2610; 170 $\mu\text{g g}^{-1}$), As (0.4–120;

3.0 $\mu\text{g g}^{-1}$), Sr (101–815; 260 $\mu\text{g g}^{-1}$), Ba (8–880; 81 $\mu\text{g g}^{-1}$), Hg (7–78730; 86 ng g^{-1}), and Pb (0.8–426; 21 $\mu\text{g g}^{-1}$).

5 | DISCUSSION

5.1 | Major elements

Calcium is largely similar in all the individuals and all the bone elements irrespective of sex, age and location (see Figure 3). This is not surprising, because Ca is a main element and it is highly metabolically controlled.

Sodium and Mg also occur in relatively high concentrations; Na varies with a factor of about two between 2200 and 5700 $\mu\text{g g}^{-1}$, while Mg varies with a factor of about five between 440 and 2500 $\mu\text{g g}^{-1}$. Even though there are some differences in the Na and Mg concentrations in individuals from the different cemeteries, Na and Mg are thought, together with Zn (which varies between 70 and 2600 $\mu\text{g g}^{-1}$), to be metabolically controlled to a substantial degree in mammals (Hyndman, 2015; Salgueiro et al., 2006). There are no indications that any of these elements is affected by diagenesis. In the present context, they merit no further attention.

5.2 | Diagenesis

Iron and Mn in archaeological bones has often been ascribed to diagenetic processes (Keeley, Hudson, & Evans, 1977; Kuczmow et al., 2010; Lopez-Gonzalez, Grandal-D'anglade, & Vidal-Romani, 2006;

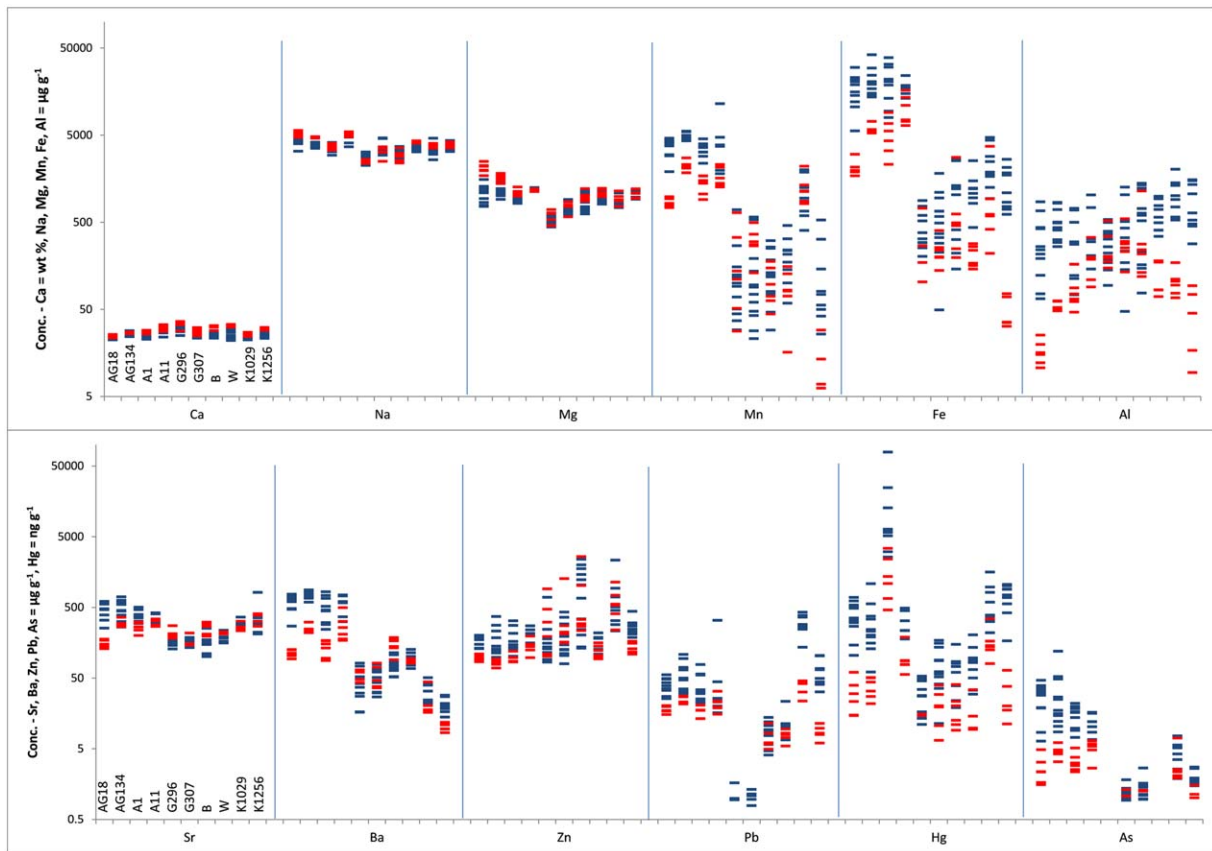


FIGURE 3 The dataset in its entirety. Blue color signifies trabecular and red cortical tissues. The y-axis scales are Ca-concentration in wt. %; Hg in ng g^{-1} , and the rest of the elements in $\mu\text{g g}^{-1}$

Parker & Toots, 1970; Skytte & Rasmussen, 2013). In non-buried bones Wallach and Chausmer (1990) reported Fe-concentrations of $51 \mu\text{g g}^{-1}$ for cortical tissue and $80 \mu\text{g g}^{-1}$ for trabecular tissue. All samples in the present study surpass these values. The elevated Fe concentrations are probably caused by diagenesis in the form of deposition of Fe^{3+} precipitated by aeration of originally anoxic groundwater, which carried iron as soluble Fe^{2+} . Regardless of the tissue, a marked difference can be seen in the Fe-concentrations between the four cemeteries. The four samples in the two cemeteries in Svendborg display highly elevated Fe concentrations compared to the other cemeteries. The trabecular samples from Svendborg have Fe concentrations varying from 5600 to $42000 \mu\text{g g}^{-1}$ and the cortical samples range between 1700 and $16000 \mu\text{g g}^{-1}$ – therefore less affected by diagenesis than the trabecular bone samples. The samples from the cemeteries at Tirup and Skt. Alberts Kapel have much lower Fe concentrations. With a single exception all the cortical samples from Tirup and Skt. Alberts Kapel have Fe-concentrations below $725 \mu\text{g g}^{-1}$, and have an average of $283 \mu\text{g g}^{-1}$, making these bones subjected to diagenesis to a lesser degree than the rest of the cemeteries. Iron concentrations are highest in the four individuals from Svendborg and the two individuals from Ribe, which cover the three cemeteries with the wettest conditions. The present data set therefore further substantiate the interpretation that the diagenetic Fe is derived from percolating groundwater, probably by precipitation of Fe^{3+} from anoxic Fe^{2+} leading to formation of pyrite framboids or oxidized framboids.

The Reference Man (1975) reported an average Mn concentration of $1 \mu\text{g g}^{-1}$ for the entire skeleton, and furthermore Mn seems to concentrate in soft tissue (Bulman, 1990). All samples in the present investigation display Mn concentrations above $10 \mu\text{g g}^{-1}$, and they are therefore most likely all affected by diagenetic Mn-deposition. Manganese has the highest concentrations in the four individuals from Svendborg and one of the individuals from Ribe (K1029), which lends further credence to Mn, like Fe, being deposited from groundwater.

Another element usually ascribed to diagenesis is Al. Aluminum in non-buried human bones has been found in concentrations up to $20 \mu\text{g g}^{-1}$ in the iliac bone (Van de Vyver & Visser, 1990). The only samples of the present study near those concentrations are the cortical tissue samples from AG18, the rest display Al concentrations 10 to 100 times higher. The likely explanation for the many elevated Al concentrations is the deposition of clay particles adhering to the lamella of the trabecular tissue or entering cracks, pores or Haversian channels in the cortical tissue; in both cases a small fraction of the particles must be escaping the decontamination procedures. Contemplating the distribution of Al in Figure 3, it can be seen that Al is more or less equally distributed in the 10 skeletons. Unlike Fe and Mn, it is therefore unlikely that Al is brought to the skeleton by groundwater.

Arsenic has previously been ascribed to diagenesis (Pike & Richards, 2002; Rasmussen, Bjerregaard, Gommessen, & Jensen, 2009). This interpretation is further substantiated by the present study.

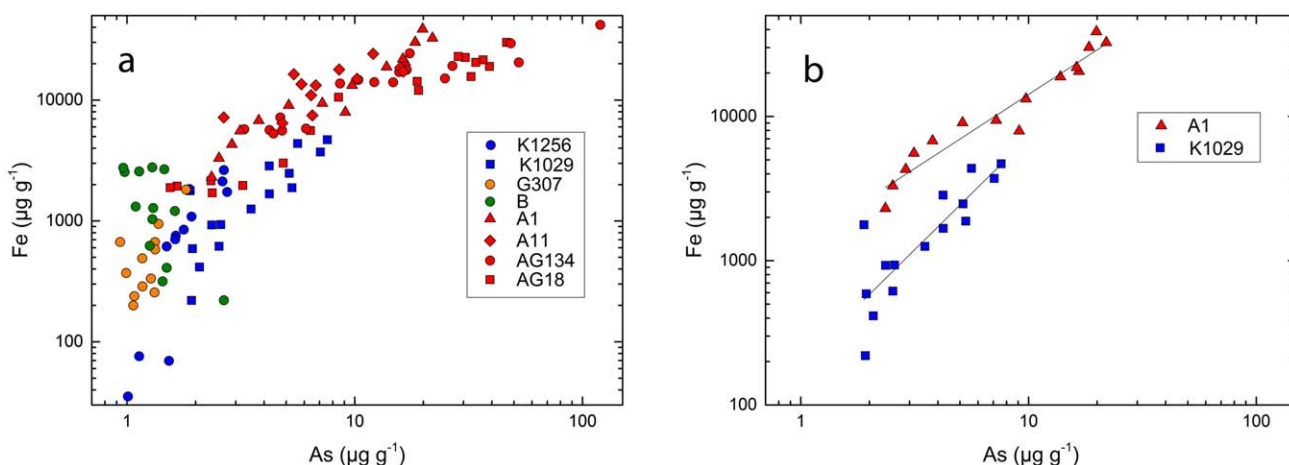


FIGURE 4 Iron as a function of As. (a) For the eight samples which yield As above LOQ; (b) a strict correlation is seen between Fe and As for some of the individuals, here shown for A1 from Svendborg and K1029 from Ribe. The straight lines are least square fits

Figure 4a shows the Fe concentration as a function of As for all the studied individuals. A clear positive correlation can be seen, although it seems that there is a trail-off in the sense that there is a limit to how high the Fe-concentration can be. The positive correlation between Fe and As can also be seen for the samples from single individuals—most pronounced for A1 from Svendborg Skt. Nicolai and Ribe K1029 (see Figure 4b). However, there are no positive correlations between Fe and As for individuals with the low As (and Fe) concentrations, for instance G307 from Tirup and B from Skt. Albert Kapel (see green and orange points in Figure 4a). As a general observation, it seems that when diagenetic deposition of Fe occurs, then As follows as well, indicating that As is also brought along with the groundwater.

The Pb concentration in trabecular femoral bone samples was observed not to be correlated with the Pb concentration in adjacent soil samples in the medieval cemetery Ole Worms Gade in Horsens, Denmark, indicating that the bones were not subject to Pb diagenesis (Rasmussen et al., 2015). Three soil samples adjacent to three skeletons have been analyzed in the present study originating from the cemetery Lindegården in Ribe. The soil samples showed Pb-concentration values of 160, 340, and 100 $\mu\text{g g}^{-1}$ and they did not correlate with the Pb concentrations in the bones, which in the present study have Pb concentrations ranging from 10 to 400 $\mu\text{g g}^{-1}$. If diagenesis were to take place, there would have to be a concentration gradient—a significant difference in concentration between the soil and the bone—to drive the transport of ions. Such gradients have, however, not been observed in any of the Danish medieval or renaissance cemeteries. These measurements augment the above mentioned findings from Ole Worms Gade in Horsens and indicate that Pb in the bones is generally not determined by diagenesis.

5.3 | Treatment of diseases

Mercury has been found to be above background levels in individuals who are diagnosed to have suffered from leprosy or treponemal infections, indicating that Hg-containing medicine was used against these

diseases in the medieval and post medieval times (Rasmussen et al., 2008, 2015; Rasmussen, Torino, Glastrup, Ramseyer, & Bjerregaard 2012; Schwarz, Skytte, & Rasmussen, 2013). In the present data, Hg is generally higher in the trabecular tissues than in the cortical (see Figure 3), as was also reported by Rasmussen et al. (2008, 2013, 2015). Two likely explanations can be mentioned for this. One is that some Hg was present in the soft tissues, which—after death—collapsed onto the large surface areas of the bone lamella. The other possibility is that these individuals may have used medication containing Hg closer to the time of death, rather than earlier in life.

5.4 | Diet and provenance

Strontium and Ba have been the elements most widely used as dietary and provenance indicators (Burton & Price, 2001; Cucina, Tiesler, Sosa, & Neff, 2011; Fabig, 2002; Schoeninger, 1979; Schutkowski, Herrmann, Wiedemann, Bocherens, & Grupe, 1999; Skytte & Rasmussen, 2013). The bulk of the present dataset is depicted in Figure 5, which shows Ba as a function of Sr.

The four locations are depicted in different colors and each individual with different symbols, but the tissue samples used are not indicated. It is apparent from Figure 5 that there is a clear separation between the four geographical areas represented by the color codes, and it is clear that the Sr-Ba scatter plot is an effective provenance indicator. However, in the present context it is furthermore interesting to look for differences between trabecular and cortical tissues. The turnover time of the bone tissue is faster in the trabecular tissues than in cortical, and therefore the cortical tissues reflect the diet/provenance earlier in life, and the trabecular later in life. During the interval between the two times, some individuals have migrated (or changed their diet), others have not. The differences can be seen in Figure 6, where Ba is plotted against Sr for three selected individuals. The closed symbols are cortical tissues, while the open are trabecular. It can be seen from Figure 6 that G307 from Tirup and K1029 from Ribe did not migrate or change their diet during these years in their respective lives.

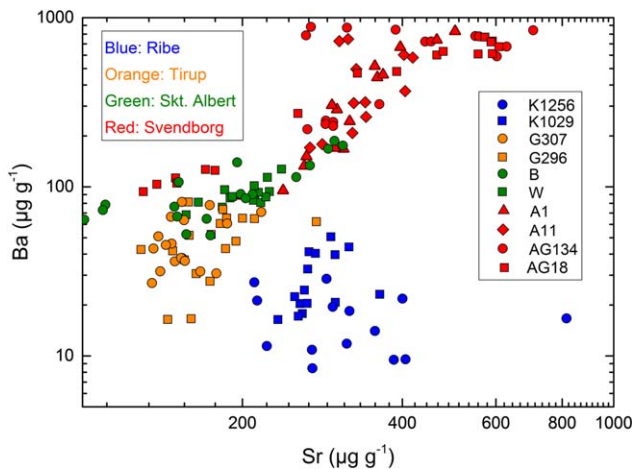


FIGURE 5 Barium as a function of Sr for the entire dataset. Color codes distinguish between location (blue: Ribe; orange: Tirup; green: Skt. Alberts Kapel; red: Svendborg). Point shapes distinguish between individuals. No distinction is made between bone elements in this plot

However, AG18 from Svendborg Franciscan Friary did indeed move or change diet in a rather drastic way. Looking at all the 10 individuals, it seems that the following individuals were stationary in the later part of their lives: K1029 and K1256 from Ribe, G307 and G296 from Tirup, W from Skt. Alberts Kapel, and A11 from Svendborg Skt. Nicolai. The following individuals had more or less pronounced migratory behavior (or dietary change) in the last part of their lives: AG18 and AG134 from Svendborg Franciscan Friary, A1 from Svendborg Skt. Nicolai Kirke, and B from Skt. Alberts Kapel.

To study the finer details between all the bone-elements, two examples of the individuals showing the largest variations, that is, the two individuals from Svendborg Franciscan Friary, are selected (see Figure 7). The dashed circles in Figure 7a and 7b encompass the cortical tissues, numbered 1 through 6. The most remote part of the distribution of the trabecular tissues are indicated by the data points to the upper right of the two diagrams. These endpoint samples are likely to have experienced the fastest turnover and the best equilibration of Sr and Ba to the new geographical location (or the new dietary habit, whichever applies). For AG134, these points encompass samples from a carpal (15), the trabecular tissue of the left tibia (18), and a tarsal (19); while for AG18, it is a sample from the sacrum (9), trabecular tissue from the right humerus (13), trabecular tissue from the left humerus (14), trabecular from the right femur (16), trabecular from the left femur (17), and trabecular from the left tibia (18).

This is interesting, because the general trend is that the fastest turnover is seen in the trabecular tissues of the long bones and the hands and the feet—that is the bones which are subjected to the highest workload and therefore the highest mechanical pressures. Macho, Abel, and Schutkowski (2005) found significant difference in trabecular wall thicknesses between “load-bearing” (foot) and “non-load-bearing bones” (hands). However, such a difference is not observed in the present study. Studying two medieval skeletons Rasmussen et al. (2013)

observed higher concentrations of Hg in the central abdomen areas than in the distal areas. In that study, it was not possible to distinguish between the two options: was the high Hg-concentration caused by the central trabecular bones having higher turnover rates? Or was it caused by postmortem deposition of Hg from inner organs like liver and kidneys onto the lamellae of the central trabecular bones? In light of the results of the present study, it becomes clear that it was probably the latter.

As noted above there can be two reasons why the Sr and Ba concentrations change through the life of an individual. One reason could be migration from one location to the other, the two locations being sufficiently apart to have different levels of bioavailable Sr and Ba in both vegetables and drinking water. Another reason could be a major change in diet of an otherwise stationary individual. In Figure 8 is shown the Ba concentration as a function of Sr concentration in modern foodstuffs, based on published analyses from predominantly UK and Canada (Canadian Total Diet Study between 1993 and 1999; Canadian Total Diet Study, 2005; Rose, Baxter, Brereton, & Baskaran, 2010; Rosenthal, 1981; Ysart et al., 1999).

Precisely how the diagram in Figure 8 translates into the skeletal system is not known, but it is obvious that a change from for example predominantly meat to predominantly shellfish can change the bioavailable Sr by about a factor of 100, and simultaneously the bioavailable Ba by a factor of about 10. Such changes in diet could very well originate from a migration from one location to another, which also most likely involved a change in the bioavailable Sr and Ba in the drinking water and local vegetables. However, it could conceivably also be caused by a change in wealth or social group, the latter in connection with marriage, for example. Some variation could also be due to changes in individual personal taste preferences, which can be driven by pure pleasure or by changes in health status. Both of these lastly

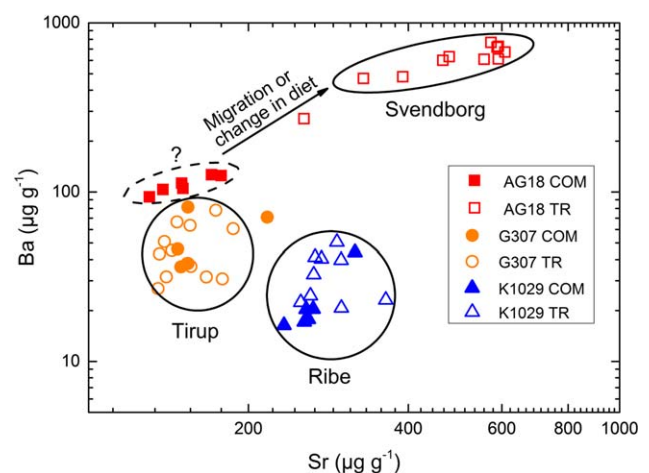


FIGURE 6 Barium versus Sr for three selected individuals. Open symbols represent trabecular samples, filled symbols cortical. It is apparent that AG18 lived at another geographical location (or alternatively changed his diet) when the cortical tissues were re-modeled and the moved to Svendborg in his later years, the chemistry of which is reflected in the trabecular tissues

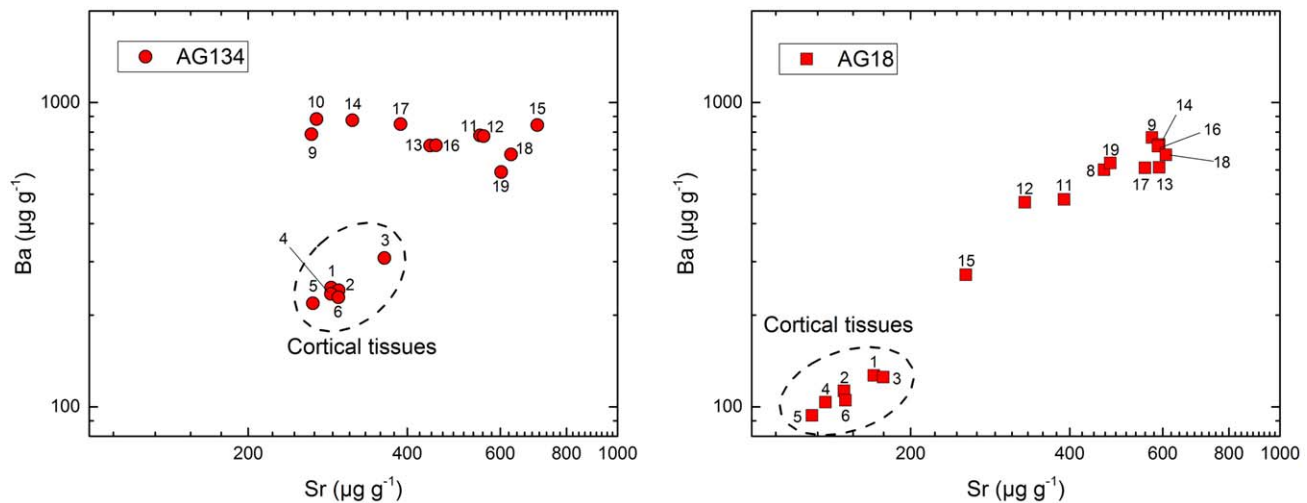


FIGURE 7 Strontium versus Ba for the two samples showing the greatest variations; AG134 and AG18 from the Franciscan Friary in Svendborg. The samples are numbered: 1: humerus right cortical; 2: humerus left cortical; 3: ulna left cortical; 4: femur right cortical; 5: femur left cortical; 6: tibia left cortical; 7: joint of sacrum; 8: vertebra; 9: sacrum; 10: ilium left; 11: rib high; 12: rib low; 13: humerus right trabecular; 14: humerus left trabecular; 15: carpal; 16: femur right trabecular; 17: femur left trabecular; 18: tibia left trabecular; 19: tarsal

mentioned changes can conceivably take place without involving any migration. All in all, it is a complicated issue.

If we look at AG18 as an example, then the change in Sr and Ba in the years between the time recorded by the cortical tissue and the later time recorded by the trabecular tissues could theoretically be due to a drastic change in diet from predominantly meat to predominantly shellfish (in this example marine foodstuffs could also be diagnosed by measuring $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Chisholm, Nelson, & Schwarcz, 1982; Schoeninger & DeNiro, 1984)). However, migrating from one location to another is also a viable and probably more likely interpretation.

One overriding concern is if the observed changes in AG18 (as an example) could instead be caused by diagenesis. To test if the drastic changes in Sr and Ba for AG18 and some of the other individuals from Svendborg could have been caused by diagenesis, the Sr and Ba concentrations have been shown as a function of the bone re-modeling turnover time in Figure 9. Although it is not a completely clear cut picture, a positive correlation can be seen between Sr-concentration and turnover time in AG18, AG134, and A1. Similarly, a positive correlation is seen between Ba-concentration and turnover time in AG18 and AG134. For A1 there is a tendency that the trabecular tissues are higher in Ba-concentrations than the cortical tissues. It should be noted that some of the noise seen in the data of these plots could be due to minor discrepancies in precisely which bones of the present investigation corresponds to the bone turnover times reported in Reference Man (1975).

As a whole, however, this is an indication that diagenesis was not responsible for the variation of Sr and Ba. As a comparison, the data for three stationary individuals have been added to Figure 9 as well—A11 from Svendborg, G296 from Tirup, and K1256 from Ribe. For these, there seems to be no correlation, which indicates the lack of diagenetic effects.

5.5 | Lead as social group indicator

Exposure to Pb in the environment translates into higher Pb concentrations in the skeleton, which has been shown for modern individuals (Erkkilä et al., 1992; Gamblin, Gordon, & Muir, 1994; Landrigan & Todd, 1994 and references therein). Similarly, Pb in archaeologically derived bones has been ascribed to exposure to Pb in the environment and food (Aufderheide et al., 1985; Qin, Li, Yang, Huang, Qin, & Xie, 2015; Rasmussen et al., 2015).

Several anthropogenic Pb-sources in medieval society can be thought of: lead coins, lead tiled roofs affecting collected drinking water, lead framed stained-glass windows, and lead-glazed ceramics.

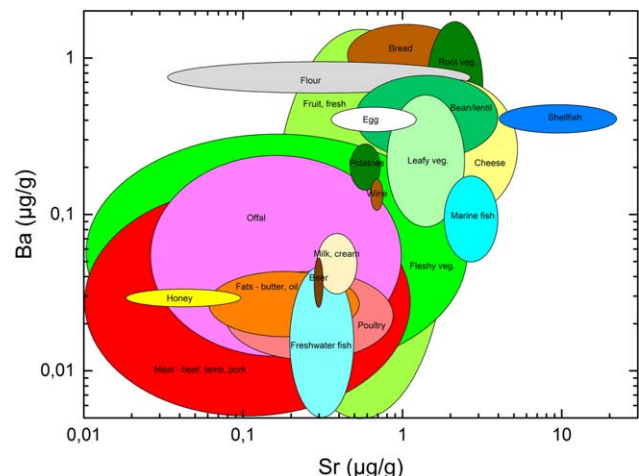


FIGURE 8 Barium versus Sr for modern foodstuff. Based mostly on Canadian and UK data (Canadian Total Diet Study between 1993 and 1999; Canadian Total Diet Study 2005; Rose et al., 2010; Ysart et al., 1999; Rosenthal, 1981)

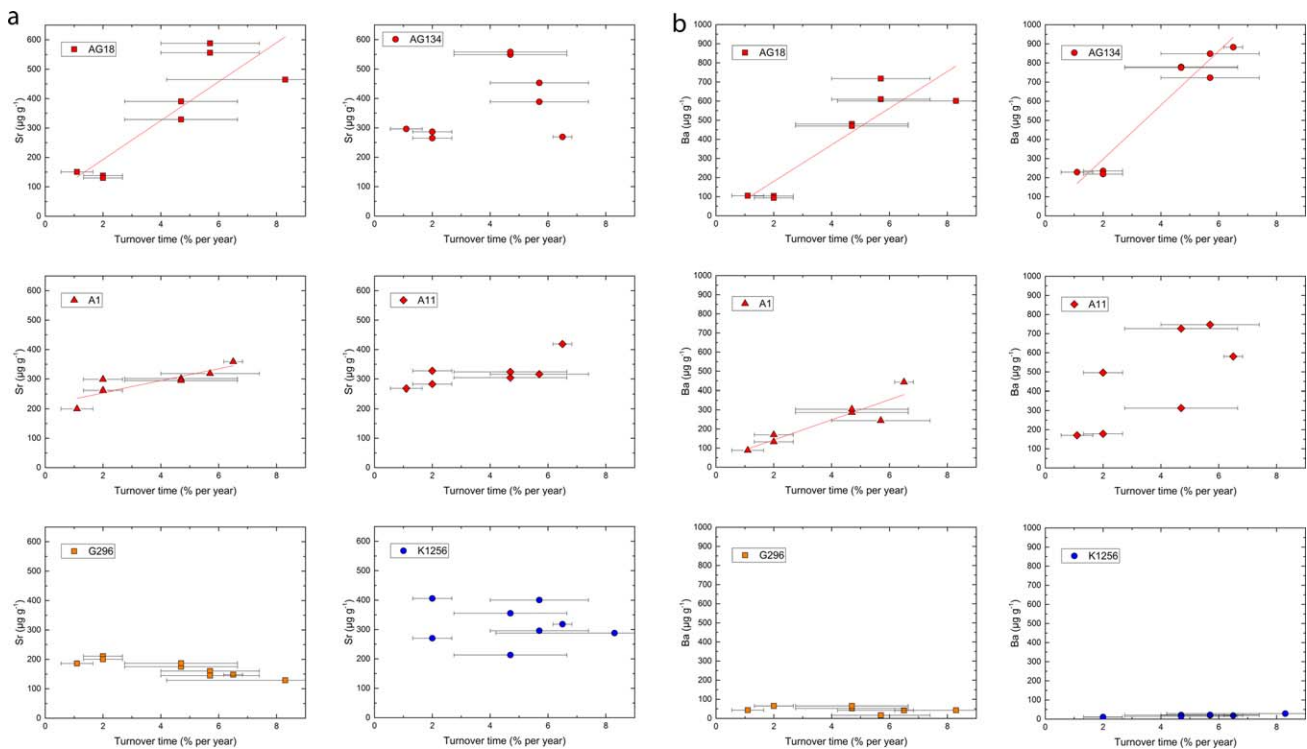


FIGURE 9 Concentrations of Strontium and Ba in $\mu\text{g g}^{-1}$ are shown as a function of turnover time in % per year. The straight line is the best linear fit. The turnover times are from Reference Man: Anatomical, Physiological and Metabolic Characteristics, ICRP (international commission on radiological protection) publication 23, by Reference Man, 1975, Amsterdam: Elsevier (p. 75), the mean turnover time in % per year and the range (in parenthesis) are: ribs: 4.7 (2.2–10); spine: 8.3 (3–6–20); ilium: 6.5 (1.2–5.6); femur cortical (shaft): 2.0 (1.5–4.2); femur trabecular (proximal end): 5.7 (2.6–9.4); and tibia cortical: 1.1 (0.4–2.6).

The latter one—the glazed ceramics—is probably the predominant source, because here acidic foodstuff got into direct contact with the PbO of the glazing, and because glazed ceramics were so widespread in Danish Medieval and Post-medieval society. Lead has been suggested as an indicator for social class by Qin et al. (2015) and Rasmussen et al. (2015). The basic argument for this hypothesis in the case of medieval Europe is that that lead-glazed ceramics were indeed the major source for Pb in medieval and post-medieval society. As shown by Rasmussen et al. (2015), Pb concentrations above background (ca., $5 \mu\text{g g}^{-1}$ for cortical tissue) are found in almost all individuals from urban communities and in ca. 1/3 of the population in rural communities. Lead-glazed ceramics were undoubtedly more expensive than plain ceramics, because of the added cost of the lead-glaze and the firewood needed for an extra firing in the kiln. Therefore, it was suggested that Pb concentrations could be interpreted as an indication of (a) high urbanization or (b) high social class in medieval society. Later, in the industrialized era, many more Pb-sources became present. In the industrialized era, the major source of Pb was again probably from food storage, namely Pb-soldered tin containers for food. In some instances, like the members of the Franklin expedition, this gave Pb bone concentrations of up to $1740 \mu\text{g g}^{-1}$ (Keenleyside, Song, Chettle, & Webber, 1996).

The data of the present investigation is depicted in Figure 10, showing the Pb-concentration as a function the Sr-concentration,

without indication the specific bone analyzed. From this plot, it can be deduced in general terms that Ribe and Svendborg are the most urbanized communities of the present selection of sites. Tirup is very low in Pb, many of the Tirup samples were below LOQ for Pb ($0.84 \mu\text{g g}^{-1}$), and Tirup is the most extreme non-migratory rural community so far excavated in Denmark (Rasmussen et al., 2015).

Conversely, the lead exposure as depicted in Figure 10 can also be viewed in another way, namely that there is a general trend of increasing lead exposure with time. Tirup is not only the most rural site but also generally the oldest (dating range ca. 1150–1350); Skt. Alberts Kapel the second oldest (ca., 1250–1536), and the selected individuals from the three cemeteries in Ribe and Svendborg the youngest. In general terms there is no doubt from the archaeological record that lead glazed ceramics became more widely used over the years from 1400 to 1700, although it is unlikely that it was widespread in the lower social classes, nor at the most rural sites at any time. Therefore, it is possible that part of the trends observed can be caused by an overlying increase in the use of lead glazed ceramics with time in Medieval and Post-medieval Denmark.

There is a marked and very interesting difference between trabecular and cortical tissues for two of the individuals, as can be seen in Figure 11, which shows the hypothetical social-class-indicator Pb as a function of the provenance-indicator Sr. Here, the cortical tissues are

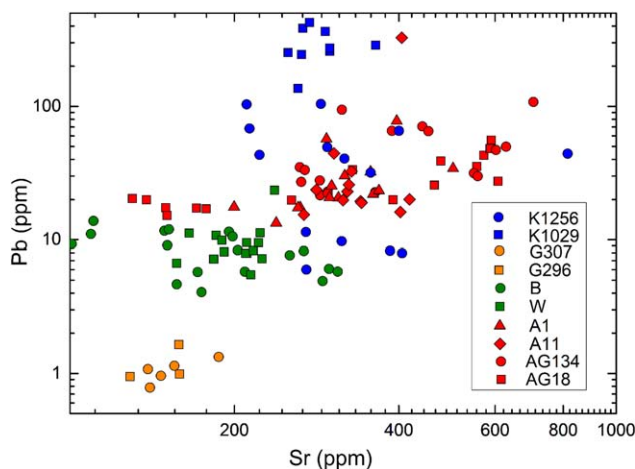


FIGURE 10 The concentration of Pb as a function of the Sr-concentration both in $\mu\text{g g}^{-1}$. The data are shown in color codes reflecting the four different locations, red: Svendborg; green: Skt. Alberts Kapel; orange: Tirup; blue: Ribe. Point shapes refer to individuals. No distinction is made to bone element

represented by closed symbols and trabecular by open symbols. The highly migratory individual AG18 from Svendborg exhibits no change in his rather low social-class-indicator Pb. Conversely, the woman K1029 from Ribe does not show any migratory behavior at all—there is a constant Sr concentration throughout, but instead K1029 shows a marked rise in Pb concentration at the time the trabecular tissue was remodeled, indicating perhaps a sudden rise in social status during this time. Such rise can be caused by many factors, to mention the most obvious: marriage, inheritance, or luck in business. It is rather interesting, that chemical analyses can shed light on such social factors in the life of an individual who lived and died many centuries ago.

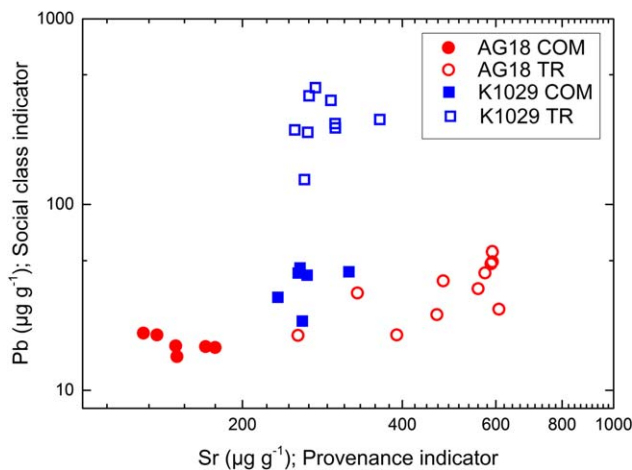


FIGURE 11 The concentration of Pb as a function of the Sr-concentration for two of the individuals, AG18 from Svendborg and K1029 from Ribe. The individuals were selected because they show drastically different behavior in the diagram. Open symbols are trabecular tissues, filled symbols are cortical tissues

6 | CONCLUSION

This is the first study of archaeologically derived medieval skeletons including as many as 19 bone samples from each skeleton. The analysis presented here substantiates previously published evidence that high As concentrations are indicative of diagenetic behavior.

The present results support that Ba- and Sr-concentrations are effective provenance indicators, which are able to distinguish between four widely separated locations in medieval Denmark. Migration behavior during the time interval reflected by the cortical tissues (ca., 5–10 years prior to death) and the time reflected by the trabecular tissues (ca., 2–3 years before death) has been observed in 4 individuals. Non-migration behavior was seen in the other six individuals.

Based on the differences in the Ba- and Sr-concentrations in all the bone samples from the two highly mobile individuals, AG18 and AG134, it is deduced that the fastest turnover is seen in the trabecular tissues of the long bones and the hands and the feet, and that these bones have a higher turnover rate than central abdomen trabecular bones, such as the ilium or spine.

Based on a scatter plot between Sr and published turnover times it is likely that the differences seen in Sr are not caused by diagenesis, but rather diet and/or provenance. Finally, Pb seems to be a viable indicator for social class and/or urbanization, with the caveat that part of the signal can be caused by an overlying increased use of lead glazed ceramics over the centuries. It would be beneficial if future research could lead to the comparison of trace element concentrations with the stable isotope ratios in many bone elements from the same individual.

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